

Algebraic, Operational, and Denotational Semantics for Quantum Communication

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Abstract

Adequacy and full abstraction connect an abstract theory to two complementary concrete techniques: an operational semantics and a categorical denotational semantics. We present a worked example for classically controlled quantum I/O, which to our knowledge has not been treated in this framework before. The abstract side is an algebraic theory $I/O \otimes \text{QUANTUM}$, presented as a tensor of parameterised algebraic theories [7]; the concrete side comprises a probabilistic operational semantics based on a notion of quantum stream, and a denotational semantics given by a free I/O monad on CP^∞ , the category of matrix algebras and CP maps with infinite biproducts, following [5]. The two concrete models are tied together by an adequacy and full abstraction theorem.

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1 Introduction

What. Classically controlled quantum communication is a programming-language primitive for sending and receiving qubits, in which *whether* a qubit is sent can depend on a classical measurement outcome. It is naturally expressed in imperative syntax — e.g. $p = \text{in}(); \text{if } (\text{measure}(p)) \text{ out}(\text{new}(|0\rangle))$ — but not in the standard quantum circuit model.

Why. Distributed quantum computing [1] is one route to scaling quantum hardware and needs send/receive primitives; textbook quantum protocols such as teleportation and BB84 already implicitly use them; and yet, prior accounts in the literature treat quantum communication mostly within quantum process calculi, where it is entangled with the additional complexity of concurrency.

How. We isolate quantum communication as an algebraic effect and study its semantics by pairing an abstract theory (operations + equations) with two concrete models, an operational and a denotational one, then connecting the two with

an adequacy + full abstraction theorem. Figure 1 summarises the structure. The talk uses this case study as a worked instance of the abstract–concrete matchings ACV is convened to study.

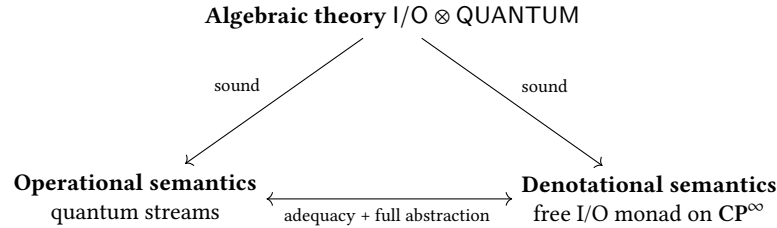


Figure 1. The abstract theory and the two concrete models. The bottom edge (Theorem 6.1) shows that the two concrete models induce the same program equivalence; both refine the abstract theory at the top.

2 Background: parameterised algebraic theories and QUANTUM

A parameterised algebraic theory (PAT) [7] is a signature of operations together with equations over their terms, whose operations carry *linear parameter contexts*: a typing judgement $\Gamma \mid \Delta \vdash c$ has Δ a list of parameter names (here, qubits) and Γ a list of continuation variables with their valences. Operations $\text{op} : (p \mid m_1 \dots m_k)$ consume p parameters and bind m_i fresh parameters in each of k continuations. Read as a type discipline, this is a linear type system for resource-sensitive effects.

Two combinators are standard: the *sum* (the free combination, no equations imposed between the two signatures) and the *tensor* (which adds the equation that every operation of one theory commutes with every operation of the other). Their constructions for the PATs of [7] are worked out in [8].

Staton’s QUANTUM [7] is a PAT for qubit quantum computation with operations $\text{new} : (0 \mid 1)$ (state preparation), $\text{apply}_U : (n \mid n)$ (unitary evolution) and $\text{measure} : (1 \mid 0, 0)$ (computational-basis measurement with classical control), together with quantum-theoretic equations such as the principle of deferred measurement. It is sound and complete with

*This presentation is about joint work with Sam Staton (Oxford).

respect to its standard operator-algebra interpretation in CP, the finite-biproduct completion of the category CPM of matrix algebras and CP maps. The analogous picture to Figure 1 is settled in this I/O-free case. We extend it to communication.

3 The algebraic theory for I/O and QUANTUM

Extending QUANTUM with $\text{in} : (0 \mid 1)$ (receive a qubit) and $\text{out} : (1 \mid 0)$ (send a qubit) lets us write classically controlled communication directly. The theory $\text{I/O} \otimes \text{QUANTUM}$ is the *tensor* (not sum) of QUANTUM with a standard I/O theory [3]. There are two design points. First, in and out do not commute with each other — the order of I/O events is part of the observable behaviour of a program. Second, in and out do commute with all of QUANTUM's operations; this is motivated by quantum theory, since for instance

$$\text{measure}(a, \text{out}(b, x), \text{out}(b, y)) = \text{out}(b, \text{measure}(a, x, y))$$

is needed to make a familiar quantum equivalence — preparing a Bell pair, sending one qubit and discarding the other equals tossing a fair coin and preparing the outcome — provable in the theory ([8], eq. 4.1.4). Tensors of theories can collapse (the Eckmann–Hilton argument for unital binary operations is the textbook example); whether $\text{I/O} \otimes \text{QUANTUM}$ collapses is ruled out *a posteriori* by the two models in §§4–5.

4 Operational semantics: quantum streams

A *quantum stream* is a $\{\text{in}, \text{out}\}^*$ -indexed family $q : \{\text{in}, \text{out}\}^* \rightarrow \text{Ob CP}$ together with CPTP transitions $T(\sigma, \text{in})$ and $T(\sigma, \text{out})$ that update a hidden environment state at each I/O event. Streams are coalgebraic in flavour; we comment on a coalgebraic recasting in §7. A configuration $\langle \rho, \sigma, c \rangle$ pairs a normalised global state ρ (over program qubits, communication qubit, and hidden state), an I/O trace σ , and a program c . The small-step probabilistic transition $\rightarrow_p^t$ is defined inductively in the program structure: measurement reduces probabilistically; other operations reduce deterministically; a continuation call terminates.

The induced *quantum equivalence* $c_1 \simeq_{qu} c_2$ holds when, for every stream and every initial state, the two programs yield the same probability-weighted distribution over (final state, continuation called, final I/O trace) triples. Terms of $\text{I/O} \otimes \text{QUANTUM}$ modulo $\simeq_{qu}$ form a model of $\text{I/O} \otimes \text{QUANTUM}$ ([8], Thm. 4.3.25): the equations of the PAT are validated by the stream dynamics.

5 Denotational semantics: a free I/O monad

The operational semantics lives in CP, which has only finite biproducts; the denotational sum over all I/O traces below needs infinitely many. Following [5], CP^∞ is the infinite-biproduct, dcpo-enriched completion of CPM, inheriting

compact closure from CPM. On CP^∞ we define

$$T(X) \triangleq \mu Y. (\text{qbit}^{\text{in}} \otimes Y) \oplus (\text{qbit}^{\text{out}} \otimes Y) \oplus X \cong \bigoplus_{\sigma \in \{\text{in}, \text{out}\}^*} \text{Aux}(\sigma) \otimes X,$$

where $\text{Aux}(\epsilon) = \mathbb{C}$ and, for $s \in \{\text{in}, \text{out}\}$, $\text{Aux}(s \cdot \sigma) = \text{qbit}^s \otimes \text{Aux}(\sigma)$; the superscripts in/out are bookkeeping, both qbit^{in} and qbit^{out} are just qbit . For every I/O trace σ , the corresponding summand tensors the qubits exchanged along σ with a value of type X .

The input case deserves a comment. An input operation morally has type $\text{qbit} \rightarrow Y$. Compact closure of CP^∞ gives $\text{qbit} \rightarrow Y \cong \text{qbit}^* \otimes Y$, and the further fact that qbit is self-dual in CPM gives $\text{qbit}^* \cong \text{qbit}$, so the input summand can be written as $\text{qbit}^{\text{in}} \otimes Y$. The two readings are reconciled by a single string-diagrammatic move: the unnormalised maximally entangled state acts as a cup, bending the input wire.

Interpreting parameter contexts as tensor powers of qbit and continuation contexts as direct sums, each term $\Gamma \mid \Delta \vdash c$ denotes a Kleisli morphism

$$[[c]] : [[\Delta]] \rightarrow T[[\Gamma]] \cong \bigoplus_{\sigma \in \{\text{in}, \text{out}\}^*, (x:n) \in \Gamma} \text{Aux}(\sigma) \otimes \text{qbit}^{\otimes n},$$

a family of CP maps indexed by (final I/O trace, continuation called) — the same indexing already used by $\simeq_{qu}$. The object $\mathbb{C}$ is a model of $\text{I/O} \otimes \text{QUANTUM}$ in Kl_T^{op} ([8], Cor. 5.4.2).

6 Adequacy and full abstraction

The two concrete models coincide on program equivalence.

Theorem 6.1 ([8], Thm. 5.3.4 (adequacy) and Thm. 5.3.6 (full abstraction)). *For all $\Gamma \mid \Delta \vdash c_1, c_2$: $[[c_1]] = [[c_2]]$ iff $c_1 \simeq_{qu} c_2$.*

This is the kind of abstract–concrete correspondence the workshop addresses: both sides are independent concrete techniques, and Theorem 6.1 shows they agree, with the abstract theory $\text{I/O} \otimes \text{QUANTUM}$ as the common interface that each refines. The proof goes via a *sum-of-paths* characterisation matching the operational sum over reduction sequences to the categorical sum over biproduct components, term by term. The matching of Theorem 6.1 does not, however, settle *completeness* of $\text{I/O} \otimes \text{QUANTUM}$: whether the equational theory derives every identification validated by the two concrete models is open ([8], Conjecture 5.4.4).

7 Discussion and open directions

The pattern *algebraic theory + two concrete models + adequacy/FA* echoes themes in the algebraic-effects programme of Plotkin and Power [6] and is instantiated, for QUANTUM alone, in [7]. The quantum I/O case is technically non-trivial because compact closure, infinite biproducts, and the possibility of a tensor-of-theories collapse each constrain the construction.

We would welcome discussion at the workshop on three open directions. (a) **Completeness** of $I/O \otimes \text{QUANTUM}$ for $\approx_{qu}$: what would a suitable axiomatic logic look like for a matching like Theorem 6.1? (b) **Coalgebraic reading**. The stream of §4 is coalgebraic, and Ceragioli et al.'s coalgebraic model of quantum bisimulation [2] appears to provide a coalgebraic framing of related phenomena; we are also curious how our denotational semantics relates to the Caus construction of Kissinger–Uijlen [4]. (c) **Algorithmic verification**. The denotational semantics is probabilistic and indexed by I/O traces; is there a decidable fragment, or a probabilistic / quantum Hennessy–Milner logic against which equivalence in $I/O \otimes \text{QUANTUM}$ could be model-checked?

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